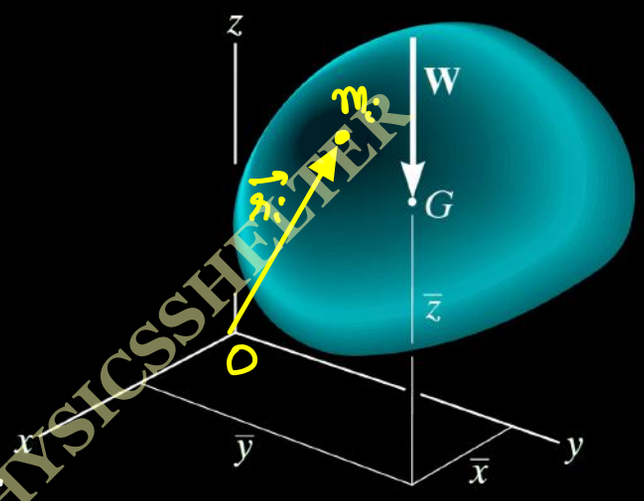
CENTRE OF MASS: The point where entire mass of a body may be assumed to be concentrated for the purpose of finding torque of its weight in a uniform gravitational field is called centre of mass. It can be shown that such a point is also the weighted average of the position vectors of all of its particles i.e. if M is the mass of the body, $\vec{r}_i$ is the position of its ith particle whose mass is m_i then, the position vector of its CM is given as:

$$\boxed{\vec{r}_{cm} = \frac{1}{M}\sum m_i \vec{r}_i} \quad (1)$$



PROOF: Torque of weight of ith particle about some point O is given as:

$$\vec{\tau}_i = \vec{r}_i \times m_i \vec{g}$$

Thus net torque of weight of all particles:

$$\vec{\tau} = \sum \vec{r}_i \times m_i \vec{g} = \left(\sum m_i \vec{r}_i\right) \times \vec{g} = M\left(\frac{\sum m_i \vec{r}_i}{M}\right) \times \vec{g}$$

$$\Rightarrow \left(\frac{\sum m_i \vec{r}_i}{M}\right) \times M\vec{g}$$, thus net torque can be found by concentrating entire mass M at point $\left(\frac{\sum m_i \vec{r}_i}{M}\right)$ therefore by definition this point is centre of mass.

→ Since equation (1) is a vector equation, it can be used to find individual x, y and z components of CM as:

$$\boxed{x_{cm} = \frac{\sum m_i x_i}{M}} \qquad \boxed{y_{cm} = \frac{1}{M}\sum m_i y_i} \text{ and } \boxed{z_{cm} = \frac{1}{M}\sum m_i z_i}$$

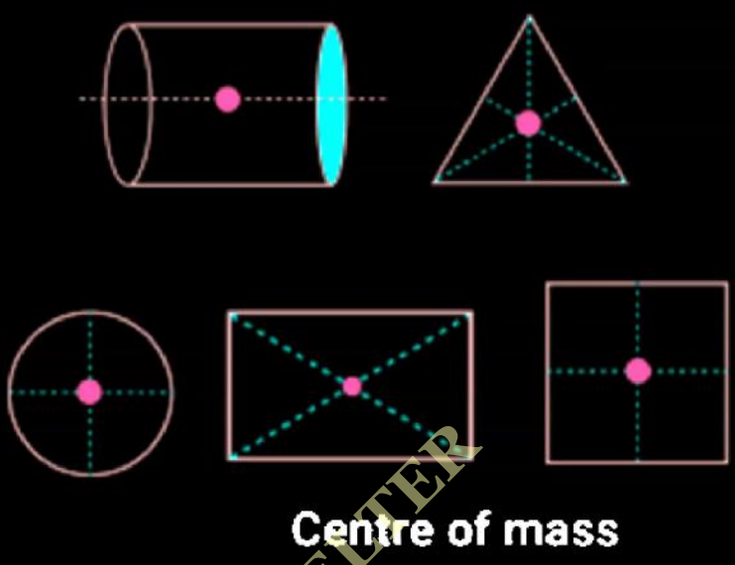
→ Centre of mass has several other interesting properties about which we will learn gradually.

→ For uniform gravitational fields centre of mass is same as centre of gravity (Hence the symbol G). For non uniform fields, we will define centre of gravity later.

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Locating G By Symmetry In Homogeneous Objects:

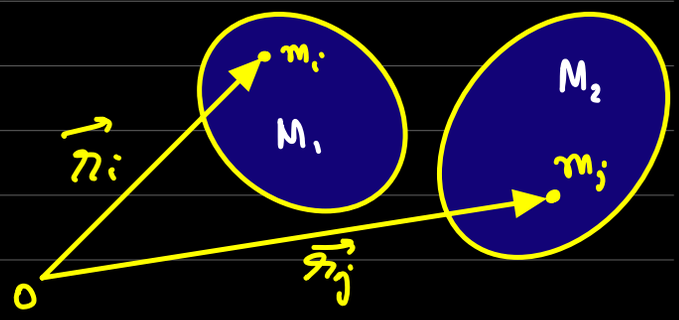
For homogeneous objects (same material density everywhere) the centre of mass can be easily located by considering intersection of planes of symmetry (3D objects like sphere, cylinder, cone etc), lines of symmetry (2D objects like circle, rectangle, isosceles triangle) or point of symmetry (for all kinds of objects where 180° opposite elements are "equally beautiful"). Some examples of locating CM by symmetry arguments are shown in the figure above.



Locating G By Composite Body Method: When two or more bodies are joined to form a composite body, for purpose of calculating G, each body may be assumed to be concentrated at its own centre of mass. This can be shown as follows:

Consider two rigid bodies as shown in adjacent figure. The combined G is given as:

$$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i + \sum m_j \vec{r}_j}{(M_1 + M_2)}$$



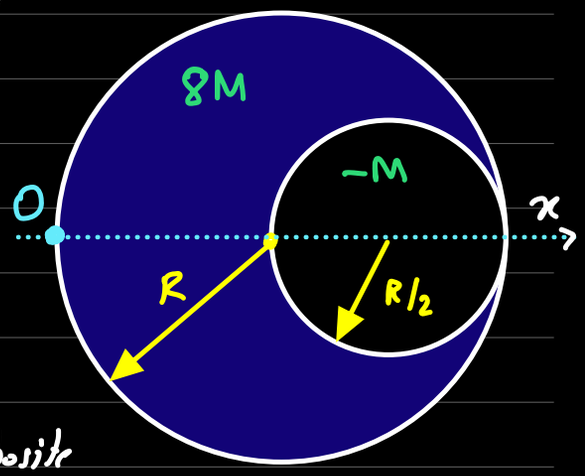
$$= \frac{M_1\left[\left(\sum m_i \vec{r}_i\right)/M_1\right] + M_2\left[\left(\sum m_j \vec{r}_i\right)/M_2\right]}{M_1 + M_2} = \frac{M_1 \vec{r}_{cm_1} + M_2 \vec{r}_{cm_2}}{M_1 + M_2}$$

QED. (Each mass is behaving like a point mass at its own centre of mass).

→ A removed volume may be considered as creating a composite with negative mass.

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Example: A sphere of radius $R/2$ is hollowed out from a solid sphere of radius R. Find the coordinate of CM of the net geometry wrt shown origin.



Solution: Let mass of removed geometry be $-M$ the mass of full sphere must be $8M$ ($\because V \propto r^3$). Thus the geometry is a composite of two spheres of masses $-M$ and $8M$. Also due to symmetry, we only need to locate the coordinate on dotted line. Now using composite body method:

$$x_{cm} = \frac{8M \times R + (-M) \times (3R/2)}{8M - M} = \frac{13R}{14} \text{ Ans.}$$

Note that since the body was made lighter on right side $\therefore$ CM shifted towards left.

Finding Centre Of Mass By Integration: Just as for discrete composite bodies, the combined CM was written as:

$$\vec{r}_{cm} = \frac{\sum M_i \vec{r}_i}{\sum M_i}$$ where $\vec{r}_i$ was the coordinate of CM of the body of mass M_i, in a similar manner, for continuous mass distributions, CM can be found by replacing summation by integration over infinitesimal elements. Thus:

$$\vec{r}_{cm} = \frac{1}{M} \int \vec{r}\, dm \quad (2)$$

→ Here dm is mass of the element and $\vec{r}$ is coordinate of CM of the element and M is the total mass.

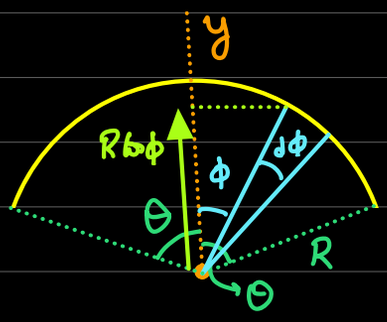
→ $\because$ (2) is a vector equation it can also be written in component form.

$$x_{cm} = \frac{1}{M} \int x\, dm \qquad y_{cm} = \frac{1}{M} \int y\, dm \qquad z_{cm} = \frac{1}{M} \int z\, dm$$

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Locating CM For Various Geometries:

CM of A Circular Arc:



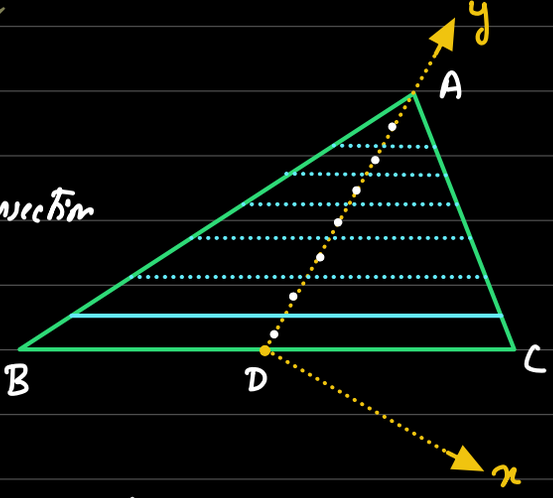
Consider the arc of angle 2θ as shown. Due to symmetry we only need to locate y coordinate of CM (∵ bisector). Now if M is mass of the arc, $dm = \frac{M}{2\theta}.d\phi$, $y = R\cos\phi$

$$\Rightarrow y_{cm} = \frac{1}{M}\int y\,dm = \frac{1}{M}\int_{-\theta}^{\theta} R\cos\phi.\frac{M}{2\theta}.d\phi = \frac{R}{2\theta}\cdot 2\sin\theta$$

$$\Rightarrow \boxed{y_{cm} = \frac{R\sin\theta}{\theta}}$$

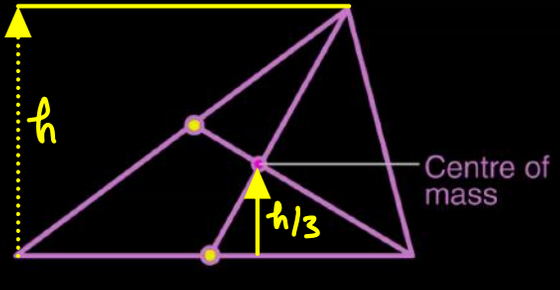
→ For semicircular arc, $\theta = \pi/2 \Rightarrow \boxed{y_{cm} = \frac{2R}{\pi}}$

CM of Triangular Lamina:



Consider a triangular lamina as shown. We will show that CM lies at intersection of medians. Consider the triangle being divided in to infinitesimal strips as shown. The CM of each of these strips lies on the median AD thus x coordinate of CM of each strip is zero. Now $x_{cm} = \frac{1}{M}\sum m_i x_i \Rightarrow 0$ ∵ each $x_i = 0$. (see figure). Thus CM must lie on the median AD. Similarly, CM must also lie on other medians therefore CM of a triangular lamina must lie at intersection of medians.

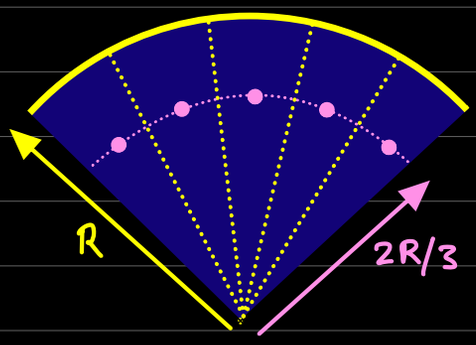
→ Since centroid divides the medians in the ratio 1:2, it can be seen from similarity that CM lies at $\frac{1}{3}$ the height of altitude on any side.



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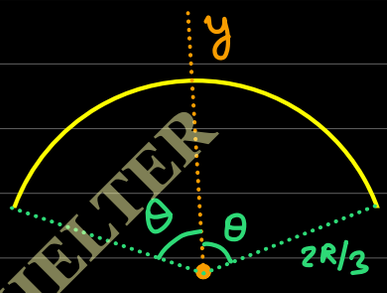
CM OF A SECTOR OF A CIRCLE :

A sector of circle can be broken in to infinite number of infinitesimal triangles with CM of each at a distance $2R/3$ from centre of curvature. Thus for purpose of calculating CM, the lamina can be replaced by arc of radius $2R/3$. Thus we can readily locate CM of a sector by just replacing R with $2R/3$ in the result for CM of an arc. Thus for sector,



$$y_{CM} = \frac{2}{3}\left(\frac{R\sin\theta}{\theta}\right)$$

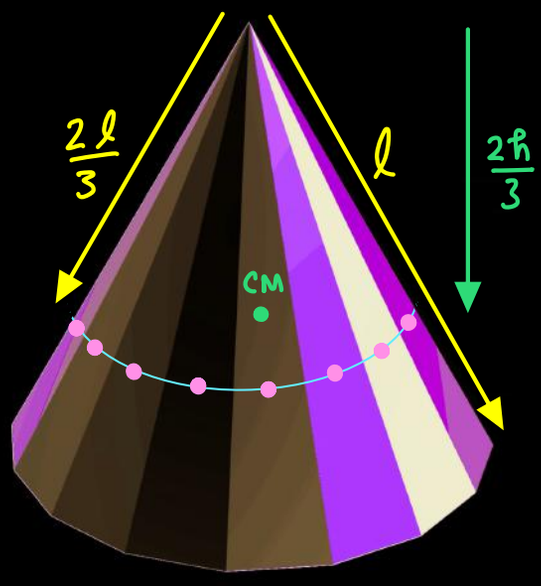
θ is semi vertex angle.



→ For semi circular lamina, $\theta = \frac{\pi}{2} \Rightarrow$ $y_{CM} = \frac{4R}{3\pi}$

CM OF A CONICAL SHELL :

A conical surface can be visualized to be made of several infinitesimal isosceles triangular strips as shown. If ℓ is the slant height of the cone, then CM of each of the triangular strips is located at a distance $\ell/3$ from the base. Thus entire cone can be replaced by a circle as shown. Using similar triangles argument, it can be readily seen that height of CM of a conical shell is $h/3$ from the base. Thus $y_{CM} = h/3$

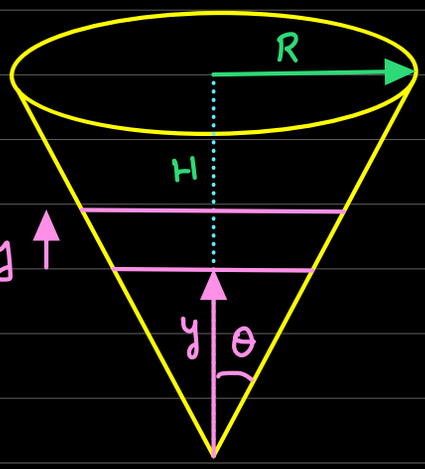


→ The same result holds if the conical shell were not right circular.

→ The same result holds for pyramid shells of all kinds.

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CM OF A SOLID CONE :



Let M be the mass of a cone of radius R and Height H (Semi-vertex angle θ).

Now consider a disc of thickness dy at a height y from the tip.

$$dm = \frac{M}{\frac{1}{3}\pi R^2 H} \cdot \pi (y \tan\theta)^2 dy, \quad \tan\theta = R/H$$

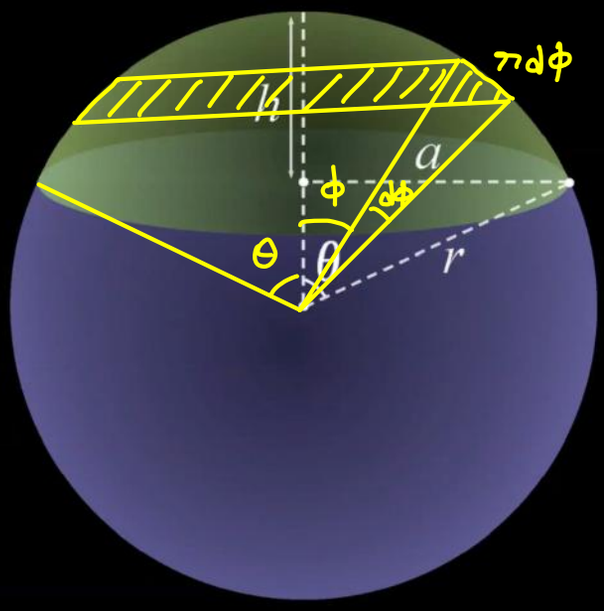
$$\Rightarrow dm = \frac{3M}{\pi R^2 H} \times \pi y^2 \times \frac{R^2}{H^2} dy = \frac{3M}{H^3} y^2 dy$$

Thus $y_{cm} = \frac{1}{M}\int y\, dm = \frac{1}{M} \times \frac{3M}{H^3} \int_0^H y^3 dy = 3H/4$. Thus, for a solid cone, $y_{cm} = \frac{3H}{4}$ from Apex or $\boxed{y_{cm} = H/4}$ from circular face

→ The same result holds even if the cone were not right circular.

→ Same result holds for all kinds of pyramids.

CM OF SPHERICAL SHELL CAP:



Consider a spherical shell cap of mass M and semi-vertex angle θ as shown.

Now, $dm = \frac{M}{2\pi r^2(1-\cos\theta)} \times 2\pi r \sin\phi \times r\, d\phi$

$$= \frac{M}{1-\cos\theta} \sin\phi\, d\phi, \quad y = r\cos\phi$$

thus, $y_{cm} = \frac{1}{M}\int_0^{\theta} \frac{M}{1-\cos\theta} \cdot r\sin\phi\cos\phi\, d\phi$

$$= \frac{r}{2(1-\cos\theta)} \int_0^{\theta} \sin 2\phi\, d\phi$$

$$= \frac{r}{4(1-\cos\theta)} \left[-\cos 2\phi\right]_0^{\theta} = \frac{r}{4}\left[\frac{1-\cos 2\theta}{1-\cos\theta}\right] = \frac{r}{4}\left[\frac{\sin^2\theta}{\sin^2\theta/2}\right]$$

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Thus for spherical shell cap, $y_{cm} = \frac{R}{4}\left[\frac{\sin\theta}{\sin\theta/2}\right]^2 = \frac{R}{4}\left[2\cos\theta/2\right]^2$

Thus $\boxed{y_{cm} = R\cos^2(\theta/2)}$ (θ is the semi-vertex angle).

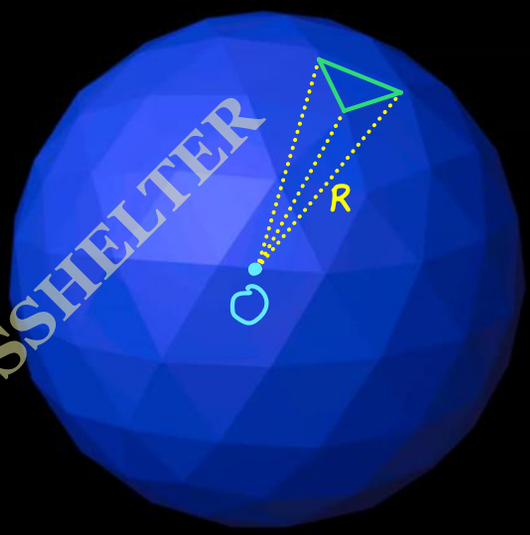
→ For hemispherical shell, $\theta = 90°$ thus $\boxed{y_{cm} = R/2}$

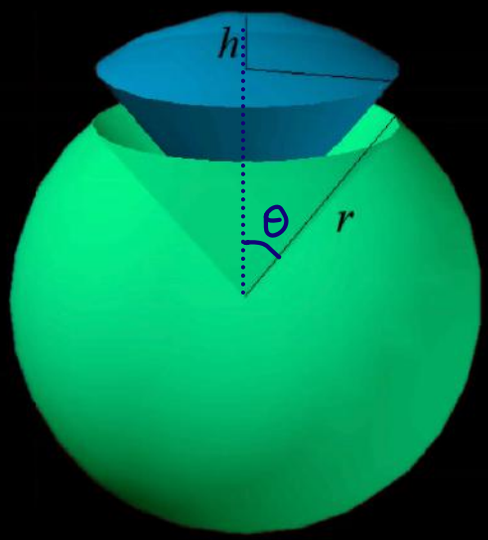
CM OF A SOLID SPHERICAL SECTOR (SPHERICAL CONE)

A solid sphere can be thought of as made of infinite number of infinitesimal pyramids (see figure for visualization). Centre of mass of each such pyramid is at a distance $3R/4$ from the centre of the sphere.

In a similar manner, a spherical sector can also be visualised as made up of several pyramids with their common tip at the centre of curvature of the sphere.

Now if each of these infinitesimal pyramids is replaced by a point mass at a distance $3R/4$ from the centre, we end up getting a spherical shell cap of radius $3R/4$ and semi vertex angle θ. Thus we can find the CM of spherical cone by simply replacing R by $3R/4$ in the previous result (for shell cap). Thus, for spherical cone,



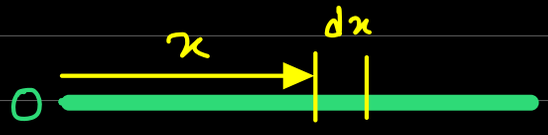


$$\boxed{y_{cm} = \frac{3}{4}R\cos^2(\theta/2)}$$

→ For a solid hemisphere, $\theta = \pi/2$ thus, $\boxed{y_{cm} = 3R/8}$

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CM OF A ROD WITH MASS DENSITY PROPORTIONAL TO x^n



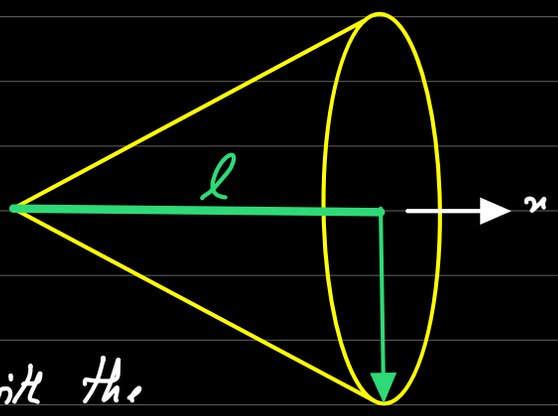
Consider a rod of length ℓ whose linear mass density varies as $\lambda = \lambda_0 \left(\frac{x}{\ell}\right)^n$

Now $x_{cm} = \frac{1}{M}\int x\,dm = \left(\int_0^\ell \lambda_0 \left(\frac{x}{\ell}\right)^n x\,dx\right) \Big/ \int_0^\ell \lambda_0 \left(\frac{x}{\ell}\right)^n dx$

Solving the integral, we get, $\boxed{x_{cm} = \left(\frac{n+1}{n+2}\right)\ell}$

PICHKOO THEOREM (FOR CM): If an object is squeezed in such a way that certain coordinate of its particles remains unchanged then CM coordinate in that direction also remains unchanged.

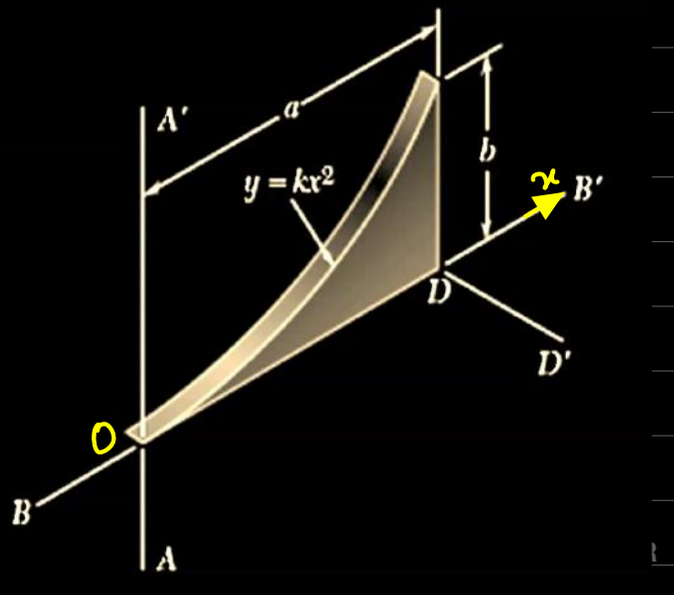
For instance if the cone in adjacent figure is squeezed to form a rod by bringing all its particles on x axis, it forms a rod of mass density $\propto x^2$, thus x coordinate of CM, $x_{cm} = \left(\frac{2+1}{2+2}\right)\ell$



i.e $x_{cm} = \frac{3}{4}\ell$ which is in agreement with the result that we derived earlier.

EXAMPLE: Locate the x coordinate of centre of mass of a parabolic spandrel as shown in the figure.

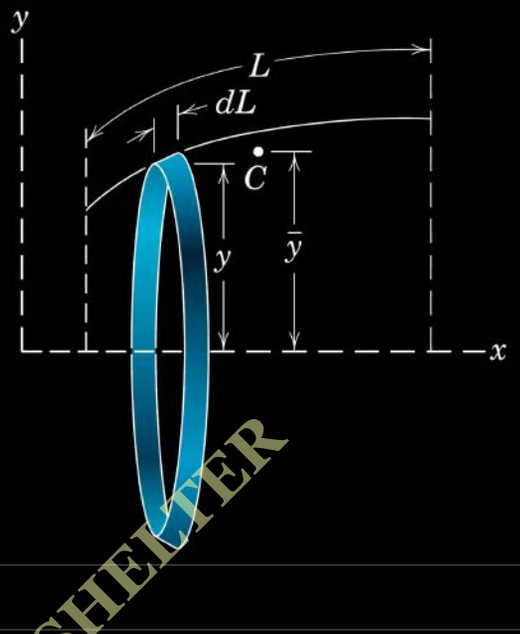
Solution: Squeezing all particles on to x axis we get a rod with mass density proportional to x^2. Thus by pichkoo theorem, $x_{cm} = \left(\frac{2+1}{2+2}\right)a = \frac{3}{4}a$ Ans.



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Theorems Of Pappus (and Paul Guldinus):

Theorem I: Surface area of a surface generated by rotating a curve in the xy plane about x axis is equal to length of the curve times distance travelled by its centre of mass. For proof see the adjacent figure. The area swept by length dL is $2\pi y\,dL$ thus total area,



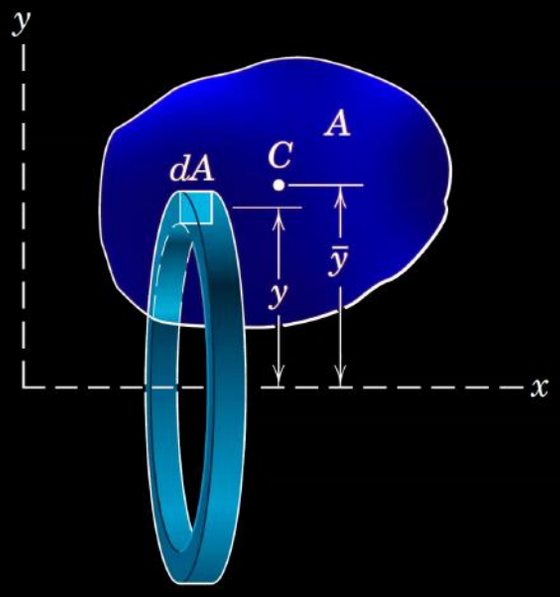
$$A = \int 2\pi y\,dL = 2\pi L\left[\frac{\int y\,dL}{L}\right]$$

$$\Rightarrow A = 2\pi \bar{y} L \quad \text{QED}$$

(Here $\bar{y}$ is y coordinate of centroid of the curve)

→ If instead of full 2π rotation, we revolved through angle θ, the surface area will be $\theta \bar{y} L$.

Theorem II: Volume of a volume generated by sweeping a laminar area in x-y plane by revolving it about x-axis is the area of the lamina times the distance traveled by its centroid. For proof see the adjacent figure. Here volume swept by area dA:



$$dV = 2\pi y\,dA \Rightarrow V = \int 2\pi y\,dA$$

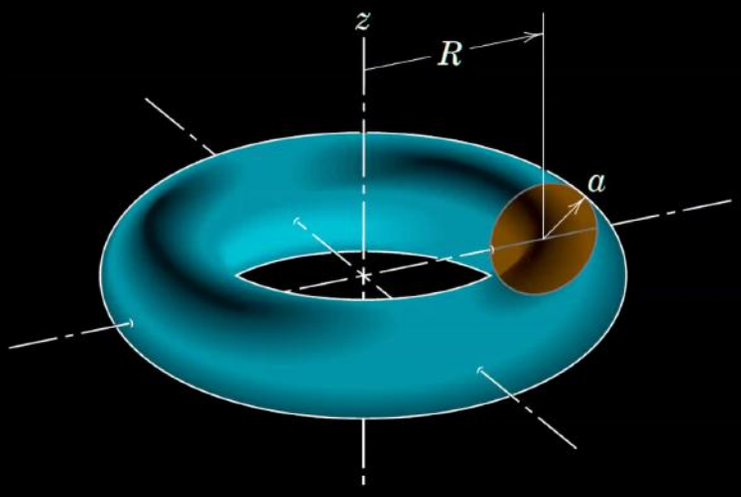
$$= 2\pi A \times \frac{1}{A}\int y\,dA = 2\pi \bar{y} A \quad \text{QED.}$$

(Here $\bar{y}$ is coordinate of centroid of the area).

→ If instead of 2π rotation, rotation is through an angle θ then $V = \theta \bar{y} A$.

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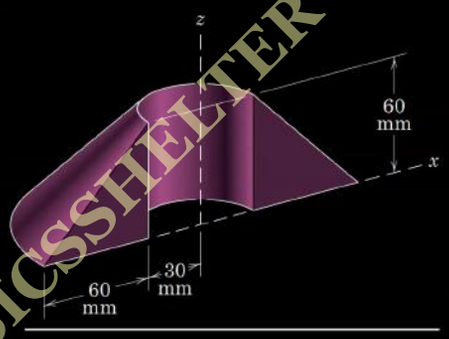
EXAMPLE: Find the volume and surface area generated by sweeping a circle as defined in adjacent figure.



Solution: $V = 2\pi R \times \pi a^2 = 2\pi^2 R a^2$ Ans.

$S = 2\pi R \times 2\pi a = 4\pi^2 R a$ Ans.

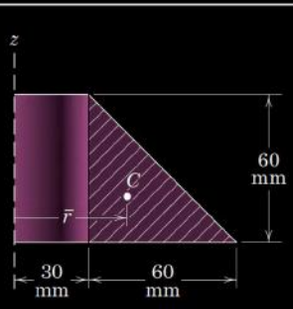
EXAMPLE: Calculate the volume generated by revolving a right triangular lamina about z axis as defined in the adjacent figure.



Solution: $V = \pi[30 + 60/3] \times \frac{1}{2} \times 60 \times 60 \text{ mm}^3$

$= 9\pi \times 10^4 \text{ mm}^3 = 90\pi \text{ cm}^3$ Ans.

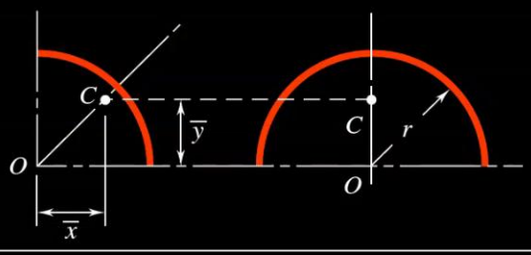
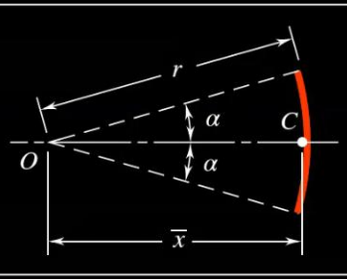
Think: How will you find surface area of the volume thus generated.



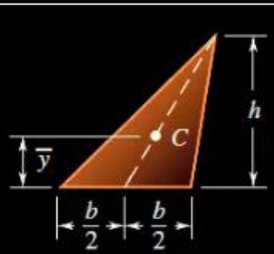
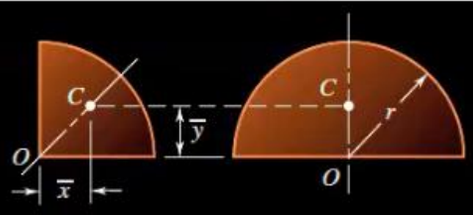
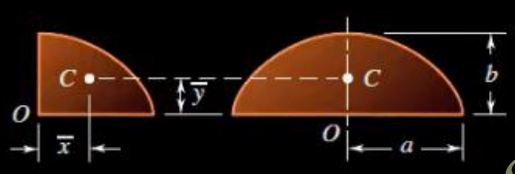
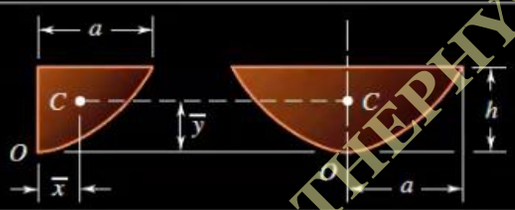
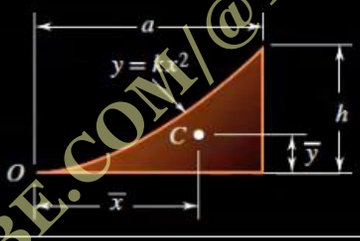
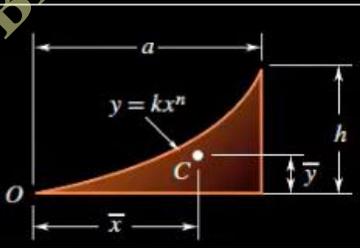
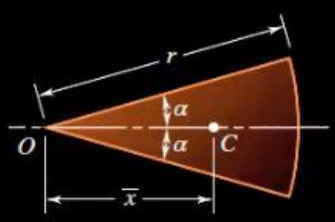
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SUMMARY OF RESULTS:

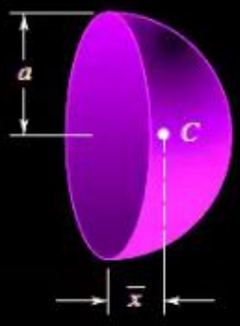
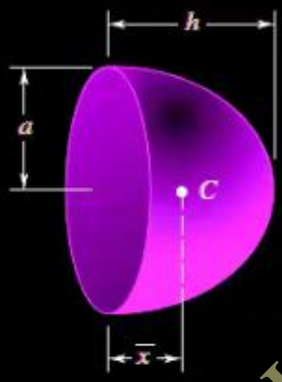
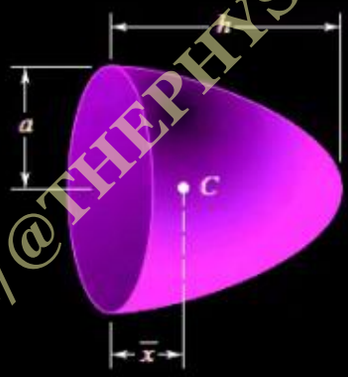
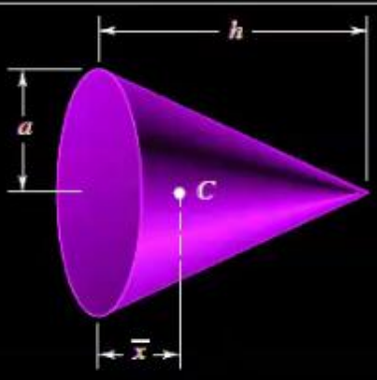
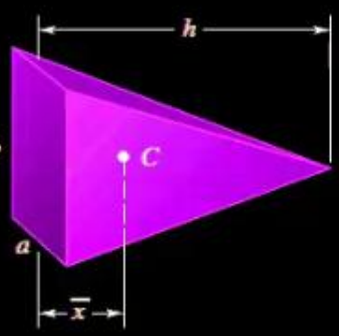
CM OF CURVES

Shape		$\bar{x}$	$\bar{y}$	Length
Quarter-circular arc		$\frac{2r}{\pi}$	$\frac{2r}{\pi}$	$\frac{\pi r}{2}$
Semicircular arc		0	$\frac{2r}{\pi}$	πr
Arc of circle		$\frac{r \sin \alpha}{\alpha}$	0	$2\alpha r$

CM OF LAMINAR BODIES:

Shape		$\bar{x}$	$\bar{y}$	Area
Triangular area			$\frac{h}{3}$	$\frac{bh}{2}$
Quarter-circular area		$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{4}$
Semicircular area		0	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{2}$
Quarter-elliptical area		$\frac{4a}{3\pi}$	$\frac{4b}{3\pi}$	$\frac{\pi ab}{4}$
Semielliptical area		0	$\frac{4b}{3\pi}$	$\frac{\pi ab}{2}$
Semiparabolic area		$\frac{3a}{8}$	$\frac{3h}{5}$	$\frac{2ah}{3}$
Parabolic area		0	$\frac{3h}{5}$	$\frac{4ah}{3}$
Parabolic spandrel		$\frac{3a}{4}$	$\frac{3h}{10}$	$\frac{ah}{3}$
General spandrel		$\frac{n+1}{n+2}a$	$\frac{n+1}{4n+2}h$	$\frac{ah}{n+1}$
Circular sector		$\frac{2r\sin\alpha}{3\alpha}$	0	αr^2

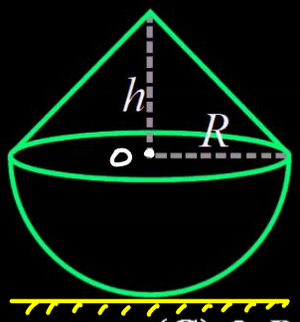
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Shape		$\bar{x}$	Volume
Hemisphere		$\frac{3a}{8}$	$\frac{2}{3}\pi a^3$
Semiellipsoid of revolution		$\frac{3h}{8}$	$\frac{2}{3}\pi a^2 h$
Paraboloid of revolution		$\frac{h}{3}$	$\frac{1}{2}\pi a^2 h$
Cone		$\frac{h}{4}$	$\frac{1}{3}\pi a^2 h$
Pyramid		$\frac{h}{4}$	$\frac{1}{3}abh$

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EXAMPLE :

A solid cone is mounted on a solid hemisphere of radius R, made of same material. What can be the height h, of the cone, so that the system is in stable configuration in the given orientation?

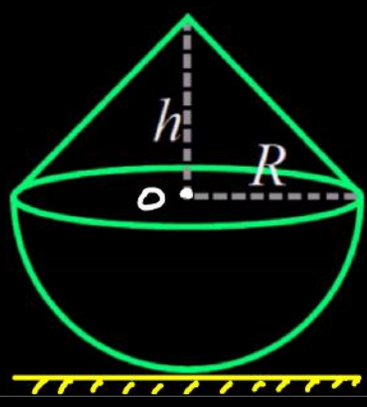


(A) 1.5 R (B) $\sqrt{3}R$ (C) 3 R (D) R

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Solution: For stability, the given configuration should have smaller potential energy than nearby configurations (upon small perturbations). Also recognize that if combined CM is at O, it will be in neutral equilibrium because will slight rolling, centre of Hemisphere will remain at same height (R) from the ground. If combined CM is below this point, the equilibrium will be stable and if it is above this point, it will be unstable. Thus in limiting case, combined CM will be at O.

Now Volume of hemisphere is $\frac{2}{3}\pi R^3$ and that of cone is $\frac{1}{3}\pi R^2 h$, thus is M is mass of hemisphere then that of cone will be $\frac{1}{2} M \frac{h}{R}$



(∵ ratio of volumes is $\frac{1}{2}(h/R)$). Now using composite body method:

$$y_{CM} = \frac{\frac{1}{2}M(h/R) \times h/4 + M(-3R/8)}{\frac{1}{2}M(h/R) + M} = 0$$

$$\Rightarrow \frac{h^2}{8R} = \frac{3R}{8} \Rightarrow h = \sqrt{3}R, \text{ thus required values, } h < \sqrt{3}R$$

thus correct options: A and D Ans.

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Motion Of Centre Of Mass:

Displacement Of Centre Of Mass: Suppose in a system of particles, i^{th} particle gets displaced through a displacement $\Delta\vec{r}_i$. Then displacement of CM can be found by subtracting initial position of CM from the final position. Thus,

$$\Delta\vec{r}_{cm} = \frac{1}{M}\sum m_i(\vec{r}_i + \Delta\vec{r}_i) - \frac{1}{M}\sum m_i\vec{r}_i$$

on $$\boxed{\Delta\vec{r}_{cm} = \frac{1}{M}\sum m_i\Delta\vec{r}_i} \quad (1)$$

Velocity Of Centre Of Mass (Constant Mass System):

By definition of CM, $\vec{r}_{cm} = \frac{1}{M}\sum m_i\vec{r}_i$, now to find velocity of CM, we can simply differentiate its position wrt time thus:

$$\vec{v}_{cm} = \frac{1}{M}\sum m_i\left(d\vec{r}_i/dt\right) = \frac{1}{M}\sum m_i\vec{v}_i, \text{ thus:}$$

$$\boxed{\vec{v}_{cm} = \frac{1}{M}\sum m_i\vec{v}_i} \quad (2)$$

Momentum Of A System In Terms Of $\vec{v}_{cm}$: (Constant Mass System)

Recognize that in equation (2), the term $m_i\vec{v}_i$ represents momentum of the ith particle thus $\sum m_i\vec{v}_i$ represents momentum of the system of particles. Rearranging (2):

$$\sum m_i\vec{v}_i = M\vec{v}_{cm} \text{ thus:}$$

$$\boxed{\vec{p} = M\vec{v}_{cm}} \quad (3) \leftarrow \text{Net momentum of a system of particles}$$

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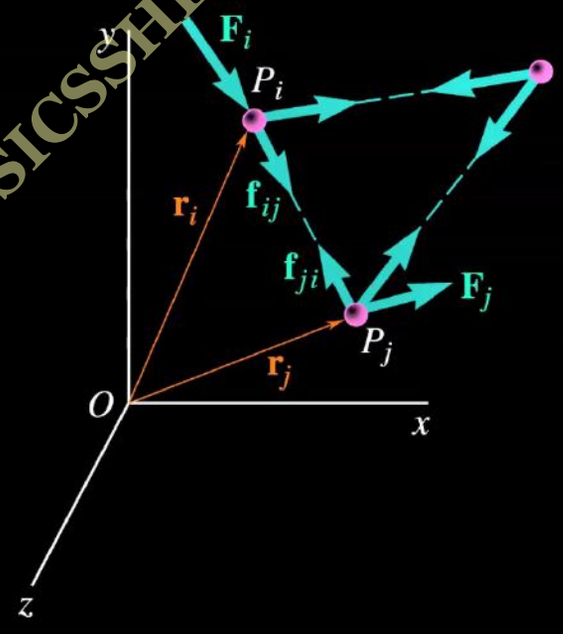
Acceleration Of Centre Of Mass (Constant Mass System):

From equation ②, $\vec{v}_{cm} = \frac{1}{M}\sum m_i \vec{v}_i$, now differentiating the equation wrt time, $\vec{a}_{cm} = \frac{1}{M}\sum m_i (d\vec{v}_i/dt)$ thus:

$$\boxed{\vec{a}_{cm} = \frac{1}{M}\sum m_i \vec{a}_i} \quad ④$$

Relation Between External Force And Acceleration Of CM:

Consider the equation ④ above, recognize that the term $m_i \cdot \vec{a}_i$ represents the force on i^{th} particle thus $\sum m_i \cdot \vec{a}_i$ represents vector sum of all the forces (both internal and external) acting on the system. Now consider two particles P_i and P_j within the system. Let $\vec{f}_{ij}$ and $\vec{f}_{ji}$ be their interaction forces (internal action reaction pairs), and let $\vec{F}_i$ and $\vec{F}_j$ be the external forces acting on them. Then, if we take the summation of forces the sum of internal forces will become zero due to their being equal and opposite (see figure). Thus summation of all the forces acting on the system is essentially the sum of external forces acting on the system. Thus rewriting ④:



$$\vec{a}_{cm} = \frac{1}{M}\sum m_i \vec{a}_i = \frac{1}{M}\left[\sum \vec{F}_i + \sum \vec{f}_{ij}\right] = \frac{1}{M}\sum \vec{F}_i = \frac{\vec{F}_{ext}}{M}$$

Here $\vec{F}_{ext}$ represents net external force on the system. Thus:

$$\boxed{\vec{F}_{ext} = M\vec{a}_{cm}} \quad ⑤ \text{ also } \boxed{\vec{F}_{ext} = d\vec{P}/dt} \quad ⑥$$

→ Net External force is equal to Mass of the System × $\vec{a}_{cm}$.

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→ Infact not only vector sum of internal forces is zero but also sum of torques of all internal forces about every point in the universe is also zero, because all action reaction pairs are contained in the same lines (equal ⊥ distances from any point). We will learn about significance of this fact when we learn about angular momentum.

LAW OF CONSERVATION OF LINEAR MOMENTUM: If net external force on a system of particles is zero then its linear momentum remains conserved. Mathematically, if $\vec{F}_{ext} = 0$ then $\vec{P}$ = Constant.

This can be shown as follows:

From previous section, $\vec{F}_{ext} = M\vec{a}_{cm}$ thus if $\vec{F}_{ext} = 0 \Rightarrow \vec{a}_{cm} = 0$

thus $\vec{v}_{cm}$ = constant, also $\vec{P} = M\vec{v}_{cm}$ thus if $\vec{v}_{cm}$ = constant then net momentum of the system must also be constant. Note that equation (5) is a vector equation thus it can also be written component-wise thus: $F_{ext_x} = Ma_{cm_x}$, $F_{ext_y} = Ma_{cm_y}$ etc.

From this we can also write law of conservation of linear momentum direction-wise, thus:

→ If net external force on a system of particles is zero in any direction, linear momentum in that direction remains conserved. Thus if $F_{ext_x} = 0 \Rightarrow P_x$ = Constant etc.

CONSERVATION OF CENTRE OF MASS (CCOM): If net external force on a system of particles is zero and the initial momentum of the system is zero then centre of mass remains fixed even if the individual particles move. This can be shown as follows:

If $\vec{F}_{ext} = 0 \Rightarrow \vec{P}$ = Constant ($\because$ COM) $\Rightarrow \vec{v}_{cm}$ = Constant.
But if $\vec{P} = 0 \Rightarrow \vec{v}_{cm} = 0$ and constantly zero thus $\vec{r}_{cm}$ = Constant.
$\Rightarrow \Delta\vec{r}_{cm} = 0$ QED.

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→ Recall that $\Delta\vec{r}_{cm} = \frac{1}{M}\sum m_i \cdot \Delta\vec{r}_i$. thus if $\vec{F}_{ext} = 0$ & $\vec{v}_{cm} = 0$

$\Rightarrow$ $\boxed{\sum m_i \cdot \Delta\vec{r}_i = 0}$

→ Since CCOM is also based on vector equations, the law can also be applied component-wise i.e if $F_{ext_x} = 0$ and $v_{cm_x} = 0$ then, $\sum m_i \Delta x_i = 0$ etc.

EXAMPLE: In a boat of mass $4M$, and length l, two men A and B of mass M and $2M$ respectively are initially standing still. The boat is also initially still. Now A travels a distance $l/4$ and B travels a distance $3l/4$ (both relative to the boat) to meet each other. Find the distance traveled by the boat on water in the process. Assume that water offers no frictional resistance.



Solution: Here, considering the x direction, $\sum F_{ext_x} = 0$ and $v_{cm_x} = 0$ thus both conditions for CCOM are met. Now let boat displace by a distance x to the right. Applying $CCOM_x$, $\sum m_i \Delta x_i = 0$ thus:

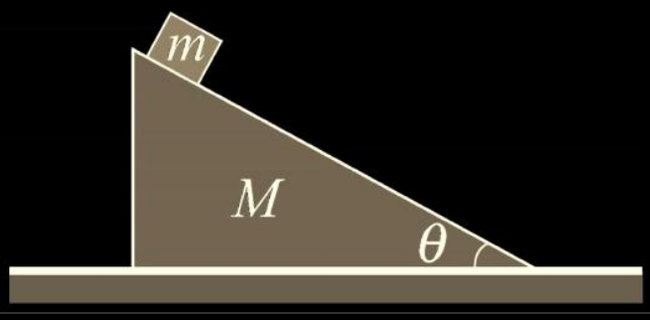
$M(x + l/4) + 2M(x - 3l/4) + 4Mx = 0$

$\Rightarrow 7Mx = \frac{5}{4}Ml \Rightarrow x = \frac{5l}{28}$, thus boat shifts $\frac{5l}{28}$ to the right.

EXAMPLE: A block of mass m starts from rest on an incline and slides a distance l relative to the incline.

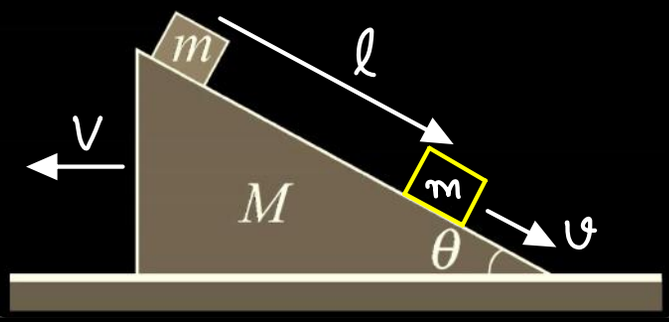
(a) Find velocity of the incline and that of the block relative to the incline after the sliding.

(b) Find displacement of the wedge in the process. (Initially wedge was stationary). All surfaces are frictionless.



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Solution: Since there is no external force in horizontal direction ∴ Horizontal momentum must be conserved. Also, since there is no kinetic friction, ∴ mechanical energy must be conserved. Let V be the velocity of wedge and v be the velocity of block relative to the wedge then:



$$MV + m(V - v\cos\theta) = 0 \quad (1) \quad (\because \text{COM})$$

Also $$mgl\sin\theta = \frac{1}{2}MV^2 + \frac{1}{2}m(V^2 + v^2 - 2Vv\cos\theta) \quad (2) \quad (\because \text{COME}).$$

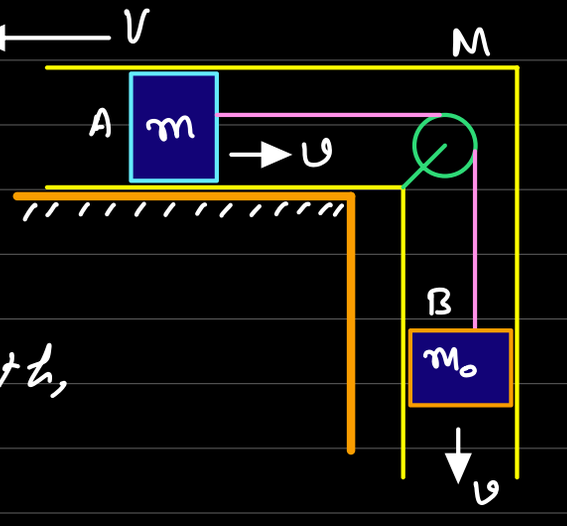
Now solving (1) & (2) $$V = \sqrt{\frac{2mgl\sin\theta\cos^2\theta}{(M+m)^2 - m\cos^2\theta(M+m)}}$$ Ans.

$$v = \sqrt{\frac{2gl\sin\theta(M+m)}{(M + m\sin^2\theta)}}$$ Ans.

(b) Now let the wedge shift a distance x to the left, then by COM, $\Sigma m_i \Delta x_i = 0$ thus: $Mx + m(x - l\cos\theta) = 0$ thus,

$$x = ml\cos\theta/(M+m)$$ Ans.

EXAMPLE: A pipe of mass M overhangs on a table as shown. The pipe has two pistons and pulley inside it arranged as shown in figure. The entire system is released from rest and there is no friction anywhere. Now after m_0 has fallen through a height h, develop a system of equations to solve for the variables (a) Leftward velocity V of the pipe (b) speed v of the pistons relative to the pipe (c) Leftward shift x in position of the pipe.



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Solution: COME: $m_0 gh = \frac{1}{2} MV^2 + \frac{1}{2} m(V-v)^2 + \frac{1}{2} m_0 (V^2 + v^2)$ ①

COM: $-MV + m(v-V) - m_0(V) = 0$ ②

CCOM: $Mx + m(x-h) + m_0 x = 0$ ③ # x, v, V.

(Now this system can be solved for variables x, v and V.

CENTRE OF MASS FRAME:

CENTRE OF MASS AS ZERO MOMENTUM FRAME: The net momentum of a system of particles as seen from the centre of mass of the system is zero. This can be shown as follows.

Velocity of ith particle as seen from CM: $\vec{v}_{icm} = \vec{v}_i - \vec{v}_{cm}$

Thus net momentum as seen from CM:

$$\vec{p}_{cm} = \Sigma m_i \vec{v}_{icm} = \Sigma m_i (\vec{v}_i - \vec{v}_{cm}) = \Sigma m_i \vec{v}_i - M\vec{v}_{cm}$$

$$= M\left[(\Sigma m_i \vec{v}_i)/M\right] - M\vec{v}_{cm} = M\vec{v}_{cm} - M\vec{v}_{cm} = 0 \quad \text{①ED.}$$

KE OF SYSTEM OF PARTICLES USING CM: Let there be a system of n particles. Now KE of ith particle: $KE_i = \frac{1}{2} m_i v_i^2$

$$\Rightarrow KE_i = \frac{1}{2} m_i \vec{v}_i \cdot \vec{v}_i = \frac{1}{2} m_i (\vec{v}_{cm} + \vec{v}_{icm}) \cdot (\vec{v}_{cm} + \vec{v}_{icm})$$

($\vec{v}_{cm}$: Velocity of CM, $\vec{v}_{icm}$: velocity of ith particle as seen from CM).

Thus $KE_i = \frac{1}{2} m_i v_{cm}^2 + \frac{1}{2} m_i v_{icm}^2 + m_i \vec{v}_{icm} \cdot \vec{v}_{cm}$, Now Net KE:

$$KE = \Sigma KE_i = \underbrace{\frac{1}{2} v_{cm}^2 \Sigma m_i}_{\text{I}} + \underbrace{\frac{1}{2} \Sigma m_i v_{icm}^2}_{\text{II}} + \underbrace{\vec{v}_{cm} \cdot \Sigma m_i \vec{v}_{icm}}_{\text{III}}$$

Now the third term in the above expression must be zero because $\Sigma m_i \vec{v}_{icm}$ is momentum in cm frame which is zero). Thus:

$$\boxed{KE = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} \Sigma m_i v_{icm}^2} = \boxed{K_{cm} + K_0}$$

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Here K_{cm} is the KE if entire mass were moving with V_{cm} i.e KE of the CM and K_0 is kinetic energy as seen from centre of mass.

→ This means that least kinetic Energy of any system of particles occurs when seen from the CM frame.

Work Done By Pseudo Forces And Uniform Gravity In CM Frame:

The total work done by pseudo forces in CM frame is zero, similarly total work done by gravitational forces in a uniform gravitational field as seen from CM frame on the particles of the system is zero.

This can be seen as follows:

Let $\vec{a}$ be the acceleration of CM, then from CM frame, power developed by pseudo force on i^{th} particle is simply, $P_i = -m_i\vec{a}\cdot\vec{v}_{icm}$ thus total power developed on all particles is:

$$P = -\vec{a}\cdot\sum m_i\vec{v}_{icm} = -\vec{a}\cdot\vec{0} \quad (\because \Sigma m_i\vec{v}_{icm} = 0, \text{ because, CM frame is a zero momentum frame})$$

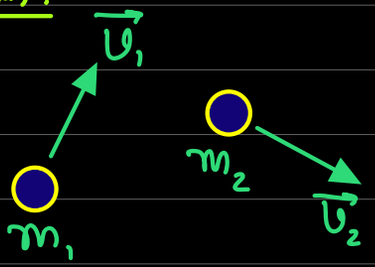
Thus net power developed by pseudo forces in CM frame is zero and therefore their net work is also zero. The proof for gravitational forces is exactly same, just that $-\vec{a}$ will be replaced by $\vec{g}$ throughout. QED.

$W_{net} = 0$ $\quad$ $W_{Pseudo} = 0$

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Two Particle System Analysis From CM Frame: Very often we come across 2 body problems in mechanics whose analysis becomes fairly easy when observed from centre of mass frame. This approach is especially useful in problems involving mechanical energy and momentum conservation. (Recall that CM frame is a zero momentum frame). Therefore in following sections we will be deriving the results for individual particle momenta and kinetic energy of the two particle systems when observed from CM frame.

Particle Momenta From CM Frame (2 Particle System):



Consider a system of two particles having masses m_1 and m_2 respectively, moving with velocities $\vec{v}_1$ and $\vec{v}_2$ respectively (see figure). Now the velocity of CM of the system: $\vec{v}_{cm} = (m_1\vec{v}_1 + m_2\vec{v}_2)/(m_1+m_2)$

Thus, momentum of particle ① in this frame:

$$\vec{p}_1 = m_1(\vec{v}_1 - \vec{v}_{cm}) = m_1\left[\vec{v}_1 - \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2}\right]$$

$$\Rightarrow \vec{p}_1 = m_1\left[\frac{m_2(\vec{v}_1 - \vec{v}_2)}{m_1+m_2}\right] = \frac{m_1m_2}{m_1+m_2}\vec{v}_{12}, \text{ thus:}$$

$$\boxed{\vec{p}_1 = \mu\vec{v}_{12}} \qquad \boxed{\mu = \frac{m_1m_2}{m_1+m_2}} \qquad \boxed{\vec{p}_2 = \mu\vec{v}_{21}} \qquad \boxed{p_1 = p_2 = p = \mu v_{rel}}$$

→ $\mu = (m_1m_2)/(m_1+m_2)$ is called reduced mass of the system because it is smaller than both m_1 and m_2 ($\because \frac{m_1m_2}{m_1+m_2} = \frac{m_1}{1+\frac{m_1}{m_2}} < m_1$)

$v_{rel} = |\vec{v}_{21}| = |\vec{v}_{12}|$.

→ Note that $\vec{p}_1$ is momentum as seen from CM while $\vec{v}_{12}$ is velocity of particle 1 as seen from particle 2 (Blunders alert!?)

Kinetic Energy From CM Frame (2 Particle System):

Recall that kinetic energy of any particle can be written as $KE = p^2/2m$. Thus from CM frame, kinetic energy of the system:

$$KE = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} = \frac{p^2}{2}\left[\frac{m_2+m_1}{m_1m_2}\right] \qquad (\because p_1 = p_2 = p = \mu v_{rel})$$

$$\Rightarrow KE = \frac{p^2}{2\mu} = (\mu v_{rel})^2 \times \frac{1}{2\mu} = \frac{1}{2}\mu v_{rel}^2, \text{ thus for two particle system, kinetic energy from CM frame: } \boxed{K_0 = \frac{1}{2}\mu v_{rel}^2}$$

→ v_{rel} is velocity of one particle relative to the other.

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Interaction Impulse Between Two Particle System (Inertial CM):

Recall that by impulse momentum theorem, $\vec{J} = \Delta\vec{p}$ thus if velocity of first particle changes relative to other particle by an amount $\Delta\vec{v}_{12}$ then the impulse acting on the particle must be $\vec{J}_1 = \mu\,\Delta\vec{v}_{12}$ and similarly $\vec{J}_2 = \mu\,\Delta\vec{v}_{21}$, thus for magnitude of impulse on each particle we can simply write:

$$\boxed{J = \mu\,|\Delta\vec{v}_{rel}|}$$

Example: Two blocks of masses m_1 and m_2 are connected with a relaxed spring on a smooth surface and then, the blocks are imparted speeds u_1 and u_2 as shown ($u_1 > u_2$). Find (a) Maximum compression in the spring (b) Maximum elongation in the spring (c) Maximum and minimum speeds of the blocks (consider possible cases).



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Solution: Here $v_{cm} = (m_1u_1 + m_2u_2)/(m_1+m_2)$. From CM frame, at maximum compression or elongation, the velocities of both blocks become zero. Thus magnitude of impulse till this event is $\mu(u_1-u_2)$, an equal impulse also acts during reformation process and reversal of speeds. Thus total impulse till reformation is $2\mu(u_1-u_2)$. If this is more than m_1u_1, m_1 must reverse its direction and its minimum speed will be zero. Now using COME from CM frame:

$$\frac{1}{2}\mu(u_2-u_1)^2 = \frac{1}{2}kx^2 \Rightarrow x = \pm\sqrt{\frac{\mu}{k}}\,(u_1-u_2)$$

Here two roots signify maximum compression and elongation. (If x is assumed compression the negative root represents elongation). Ans (a) & (b).

(c) Speed of m_2 keeps increasing until the spring remains in compression. During this time total Impulse of spring force is $2\mu(u_1-u_2)$ (∵ from CM frame momenta reverse by the

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time half oscillation cycles (completes). Thus for m_2 at max speed be v_{2max} then $m_2(v_{2max} - u_2) = 2\mu(u_1 - u_2) \Rightarrow$

$$v_{2max} = \frac{2\mu(u_1 - u_2)}{m_2} + u_2 \quad \text{Ans.}$$

For block 2, initial speed itself must be the minimum speed, because after this there is pushing impulse for half cycle and pulling impulse for next half cycle of oscillation. The pulling cycle follows the pushing cycle thus it cannot reduce the speed beyond the initial speed. Thus, $v_{2min} = u_2$ Ans.

From CM frame initial KE is maximum since PE is minimum. Thereafter the speeds oscillate periodically. Thus for particle (1), speed limits are:

$$V_{cm} \pm u_{1cm} \quad \text{and for particle 2 it is } V_{cm} \pm u_{2cm}.$$

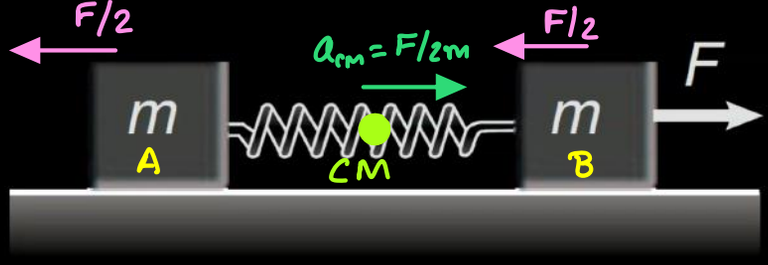
Also $u_{1cm} = \mu(u_1 - u_2)/m_1$ and $u_{2cm} = \mu(u_1 - u_2)/m_2$ ($\because v = p/m$). Now if $u_{1cm} > V_{cm}$ then the minimum speed will become zero for block 1, otherwise min speed is $V_{cm} - u_{1cm}$ i.e

$$v_{1min} = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2} - \frac{m_2(u_1 - u_2)}{m_1 + m_2} \quad \text{and} \quad v_{1max} = u_1.$$

EXAMPLE: Two blocks A and B are connected by a relaxed spring on a smooth horizontal surface as shown. Now a constant force F is applied on B. Find (a) Maximum elongation in the spring during subsequent motion (b) Maximum speeds of the blocks as seen from CM frame.



Solution: From CM frame, pseudo forces will act on the blocks (shown in pink in adjacent figure).



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Now if x_m is max elongation, from CM frame, both blocks must get displaced by $x_m/2$ in opposite directions at the time of max elongation. Now applying work energy theorem in CM frame, $F \times x_m/2 = \frac{1}{2} k x_m^2$ ($\because$ net work of pseudo forces is zero).

Thus $x_m = F/k$ Ans. (Max elongation).

Now at maximum speeds of the blocks, their acceleration must be zero. Thus if x_0 is elongation at max speeds then: $kx_0 = F/2$ $\Rightarrow x_0 = F/2k$ (1). Now by this instant, both blocks must have shifted $x_0/2$ in opposite directions. Thus:

(CHIME): $F(x_0/2) = \frac{1}{2} k x_0^2 + \frac{1}{2} m v_m^2 \times 2$ (2)

Solving (1) & (2), $v_m = \frac{F}{\sqrt{8km}}$ Ans.

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EXAMPLE: IRODOV: 1.155:

Two identical buggies move one after the other due to inertia (without friction) with the same velocity $\mathbf{v}_0$. A man of mass m rides the rear buggy. At a certain moment the man jumps into the front buggy with a velocity $\mathbf{u}$ relative to his buggy. Knowing that the mass of each buggy is equal to M, find the velocities with which the buggies will move after that. (Ignore gravity and assume all velocities as horizontal).

Solution: Considering the system of rear buggy + man, until just after the jump, CM is inertial frame. Thus impulse imparted to the rear buggy during jumping process, $J = \mu |\Delta \vec{v}_{m/1}|$. Initially relative velocity of man wrt buggy is zero and finally just after the jump, it is u. Thus impulse imparted in horizontal direction, $J_1 = \left(\frac{mM}{M+m}\right) u$, thus immediately after the jump, velocity of man is $v_m = v_0 + \frac{Mu}{M+m}$ ($\because \Delta v = J/m$).

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Now velocity of man Buggy, $v_{B_1} = v_0 - \frac{mu}{M+m}$ Ans.

Now relative to front buggy, man has a velocity $Mu/(M+m)$. Considering man + front buggy as the system, there is no external force in the horizontal direction. Thus if v_{B_2} is final velocity of the front Buggy, by momentum conservation:

$$(M+m)\, v_{B_2} = M v_0 + m\left[v_0 + \frac{mu}{M+m}\right]$$

$$\Rightarrow \quad v_{B_2} = v_0 + \frac{Mmu}{(M+m)^2} \quad \text{Ans.}$$

EXAMPLE: IRODOV: 1.156:

Two men, each of mass m, stand on the edge of a stationary buggy of mass M. Assuming the friction to be negligible, find the velocity of the buggy after both men jump off with the same horizontal velocity **u** relative to the buggy: (1) simultaneously; (2) one after the other. In what case will the velocity of the buggy be greater and how many times?

Solution: Change in velocity depends on specific impulse (i.e impulse per unit mass). Now when both men jump together, impulse = μu_{rel}

$$\Rightarrow \quad J = \frac{2mMu}{M+2m}, \text{ thus specific impulse on Buggy: } \frac{J}{M} = \frac{2mu}{M+2m} = V$$

In second case, when the first man jumps, specific impulse on Buggy + remaining man system, $\frac{J_1}{M+m} = \left[\frac{(M+m)m}{(M+2m)} \times u\right] \cdot \frac{1}{(M+m)}$

$\Rightarrow$ specific impulse, $V_1 = \left(\frac{mu}{M+2m}\right)$, now when second man jumps,

specific impulse on Buggy: $V_2 = \left(\frac{Mmu}{M+m}\right) \times \frac{1}{M} = \left(\frac{mu}{M+m}\right)$

Thus cumulative specific impulse: $V_1 + V_2 = V' = \frac{mu}{M+m} + \frac{mu}{M+2m}$

$V < V'$ thus specific impulse is greater in second case and therefore buggy has greater speed in second case.

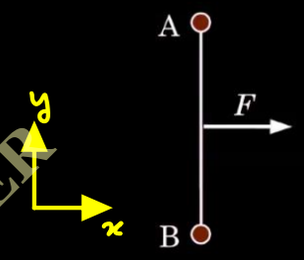
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Now ratio of speeds : $\frac{v'}{v} = \left[\left(\frac{mu}{M+m}\right) + \frac{mu}{M+2m}\right] \Big/ \left(\frac{2mu}{M+2m}\right)$

$= \frac{1}{2}\left[1 + \frac{M+2m}{M+m}\right] = \frac{1}{2}\left[1 + 1 + \frac{m}{M+m}\right] = 1 + \frac{m}{2(M+m)}$ Ans.

Example: Pathfinder: Build-3 (WPE):

Two small identical discs each of mass $m = 1.0$ kg placed on a frictionless horizontal floor are connected by a light inextensible cord of length $l = 1.0$ m. Now the midpoint of the cord is pulled perpendicular to the line joining centres of the discs by a constant force $F = 2.0$ N. What is velocity of approach of the discs, when they are about to collide?

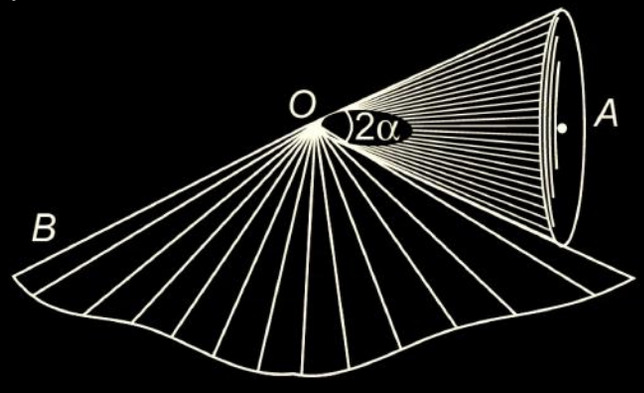


Solution: From CM frame, The balls just appear to approach each other along the y axis and pseudo force do no work. Also the mid point of the cord advances through a distance $l/2$. Thus if v is the approach velocity of balls from CM frame just before the collision, using WET:

$$F\,l/2 = \frac{1}{2}\mu v^2 = m/4\, v^2 \Rightarrow v = \sqrt{2Fl/m}$$ Ans.

Example: Irodov: 1.146:

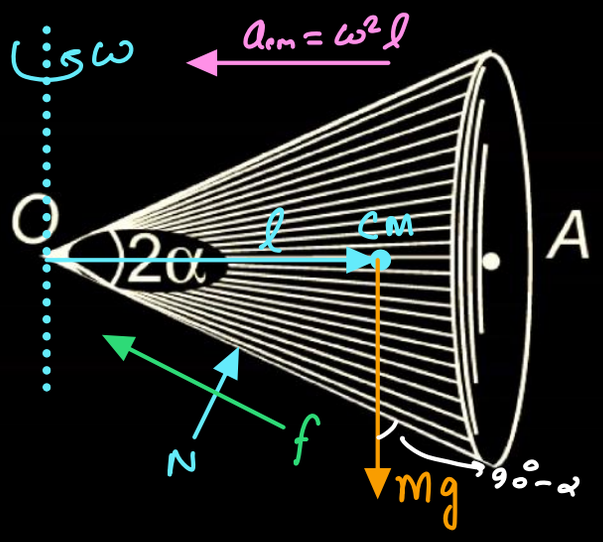
A round cone A of mass m and half angle α rolls without slipping along a round conical surface B so that its apex O remains stationary. The centre of gravity of the cone is at same level as O and at a distance l from it. The cone's axis moves with angular velocity ω. Find (a) Static friction acting on the cone (b) Maximum value of ω to roll without skidding if coefficient of friction between surfaces is μ.



Solution: Consider the FBD of the cone as shown in the figure. The acceleration of CM is $\omega^2 l$ towards the apex.

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Now applying $\vec{F}_{ext} = M\vec{a}_{cm}$ along the slant height and $\perp$ to slant height:



$$f - mg\sin\alpha = m\omega^2 l\cos\alpha \Rightarrow$$
$$f = mg\left[\sin\alpha + \frac{\omega^2 l}{g}\cos\alpha\right] \text{ Ans.}$$

Also $mg\cos\alpha - N = m\omega^2 l\sin\alpha$

$$\Rightarrow N = mg\left[\cos\alpha - \frac{\omega^2 l}{g}\sin\alpha\right]$$

At max ω, $f = \mu N \Rightarrow \sin\alpha + \frac{\omega^2 l}{g}\cos\alpha = \mu\left[\cos\alpha - \frac{\omega^2 l}{g}\sin\alpha\right]$

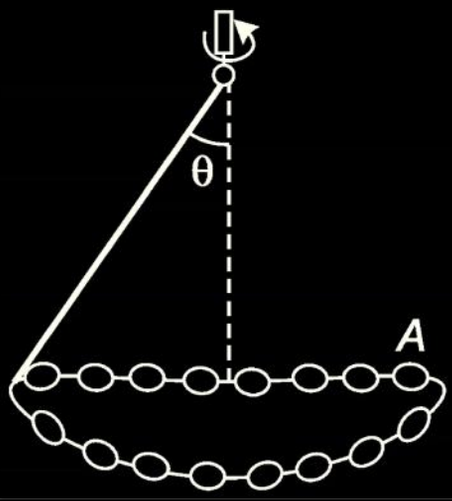
$$\Rightarrow \frac{\omega^2 l}{g}\left[\cos\alpha + \mu\sin\alpha\right] = \mu\cos\alpha - \sin\alpha \Rightarrow \omega = \sqrt{\frac{g}{l}\left[\frac{\mu\cos\alpha - \sin\alpha}{\cos\alpha + \mu\sin\alpha}\right]} \text{ Ans.}$$

or permissible ω: $\omega < \sqrt{\frac{g}{l}\left[\frac{\mu - \tan\alpha}{1 + \mu\tan\alpha}\right]}$ Ans.

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EXAMPLE: IRODOV: 1.145:

A closed chain A of mass $m = 0.36$ kg is attached to a vertical rotating shaft by means of a thread and rotates with constant angular velocity $\omega = 35$ rad s^{-1}. The thread forms an angle $\theta = 45°$ with the vertical. Find (a) Distance of centre of gravity of chain from rotation axis (b) Tension in the thread.



Solution: $\vec{F}_{ext} = M\vec{a}_{cm}$. Now suppose cm is at a distance x from rotation axis, thus applying Eom for the system of particles (i.e $\vec{F}_{ext} = M\vec{a}_{cm}$):

$$T\cos\theta = mg \quad (1)$$

and $$T\sin\theta = m\omega^2 x \quad (2)$$

Solving (1) & (2),

$$x = g\tan\theta/\omega^2 = 0.8\text{cm}, \quad T = mg/\cos\theta = 5N \text{ Ans.}$$

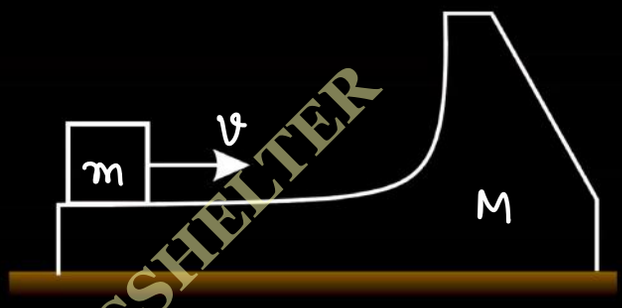
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"NINJA TEKNIK FRAME" ☺: Sometimes in problems CM is accelerated and hence non-inertial, in such cases it is often convenient to work from an inertial frame moving with initial velocity of CM. This usually simplifies the Energy conservation equation and the momentum conservation equations.

EXAMPLE : IRODOV: 1.163:

A body of mass M with a small disc of mass m placed on it rests on a smooth horizontal surface. The disc is given a speed v as shown. To what height (relative to initial level) will the disc rise?



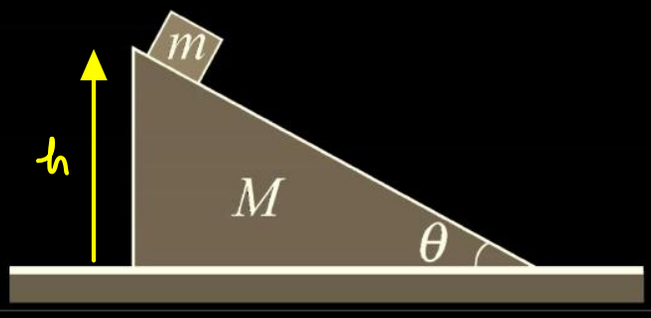
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Solution: Here the combined CM will have vertical acceleration so it will be a non-inertial frame. So let us try working from "NTF": which is inertial. In such a frame horizontal momentum is always zero so we don't need to write momentum equation. At the highest point, the velocity of CM is zero from the "NTF". (Horizontal velocity of CM was always zero from this frame). Now let us apply conservation of mechanical energy from our frame. Let h be the maximum height above initial level then:

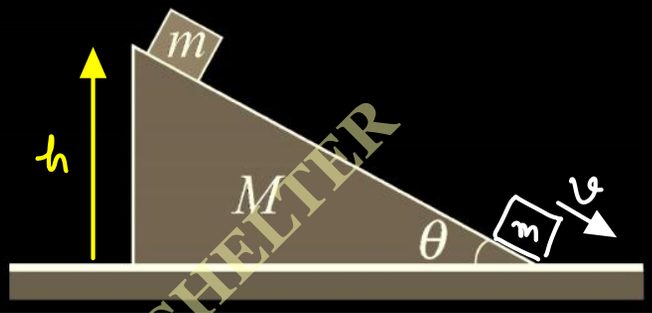
$$\frac{1}{2}\mu v^2 = mgh \Rightarrow h = \frac{Mv^2}{2(M+m)g} \quad \text{Ans.}$$

→ Appreciate the beauty of "NTF". Just one equation and we get the answer in just one line.

EXAMPLE: The system is released from rest from the shown configuration. Find the velocity of the block relative to the wedge when it reaches the bottom. (All surfaces are frictionless).



Solution: Here ground frame itself is the NTF. Here the CM has only vertical velocity. Now recall that KE of any system: $KE = K_{cm} + K_0$, thus after falling through a distance h if the speed of block relative to the wedge is v, then by energy conservation:



$$mgh = \frac{1}{2}\mu v^2 + \frac{1}{2}(M+m)v_{cm}^2$$

Now $v_{cm} = \frac{mv\sin\theta}{(M+m)}$ (∵ vertical velocity from ground frame and wedge frame are same for the block)

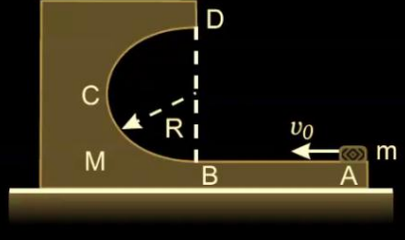
Thus:

COME: $$mgh = \frac{1}{2}\frac{Mm}{(M+m)}v^2 + \frac{1}{2}(M+m)\left[\frac{mv\sin\theta}{M+m}\right]^2$$

$$\Rightarrow gh = \frac{1}{2}v^2\left[\frac{M+m\sin^2\theta}{M+m}\right] \Rightarrow v = \sqrt{2gh\left(\frac{M+m}{M+m\sin^2\theta}\right)}$$

→ Here also we could write COME equation using a single variable using the formula $KE = KE_{cm} + KE_0$

EXAMPLE: Figure shows a small block of mass $m = 1$ kg which is given a horizontal velocity $v_0 = 10$ m/s on the horizontal part of the bigger block of mass $M = 9$ kg placed on a horizontal floor. The curved part of the surface shown is semicircular of radius $R = 1$ m. Find the distance from point B where the block m lands finally after looping the semicircular part BCD. Neglect friction everywhere. Assume that the horizontal portion AB is long enough. ($g = 10$ m/s^2)



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Solution: Let us work from NTF. When the block reaches point D, let relative velocity of the smaller block be v. Now by applying COME in the NTF:

$$\frac{1}{2}\mu v_0^2 = \frac{1}{2}\mu v^2 + 2mgR \quad \Rightarrow \quad v = \sqrt{v_0^2 - \frac{4mgR}{\mu}}$$

$$\Rightarrow v = \sqrt{v_0^2 - \frac{4(M+m)gR}{M}}$$

Now time of flight after leaving D, $T = \sqrt{\frac{4R}{g}}$, thus range,

$$\ell = vT = \sqrt{\frac{4R}{g}\left(v_0^2 - \frac{4(M+m)}{M}gR\right)} \text{ Ans.}$$

→ Think: What is the radius of curvature of the block's path at highest point from the NTF and CM.

Solution: At highest point, the groove is an inertial frame ∴ vertical acceleration from NTF is same as that seen from the groove i.e v^2/R. However from NTF, the velocity of the block is $\mu v/M$. Thus radius of curvature is $(\mu v/m)^2 / (v^2/R) = \left(\frac{M}{M+m}\right)^2 R$ Ans.

From CM frame, at the highest point, speed of block is same as that seen from NTF; however acceleration of CM is $\left(\frac{mv^2}{R}\right)/(M+m)$

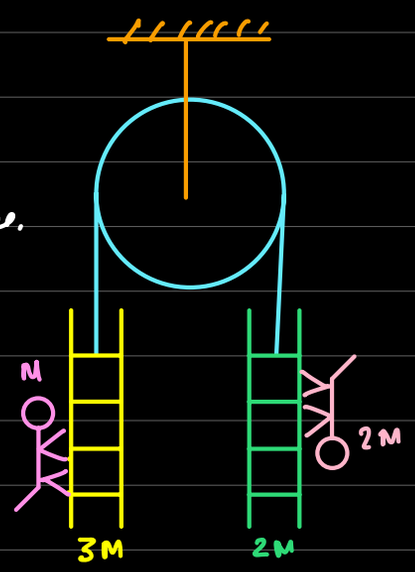
Thus vertical acceleration of block as seen from CM is $\frac{v^2}{R} - \frac{mv^2}{R(M+m)}$

i.e $a_c = \frac{Mv^2}{R(M+m)}$ thus radius of curvature $(\mu v/m)^2 / (Mv^2/R(M+m))$

$= \left(\frac{M}{M+m}\right)R$ Ans.

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EXAMPLE: Two monkeys of masses M and 2M are sitting still on ladders of masses 3M and 2M as shown in the adjacent figure. Now M climbs up a distance ℓ relative to the ladder and 2M climbs down by same distance relative to the ladder. By how much distances do the ladders shift in the process?



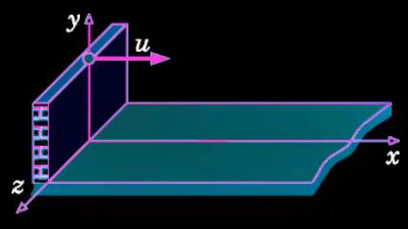
Solution: If we consider monkey + ladder as the system, we immediately recognize that net external force (TD 4Mg) are identical on LHS and RHS of the pulley. Also since total mass on both sides is equal $\therefore a_{cm}$ is equal and since initial v_{cm} is zero on both sides $\therefore v_{cm}$ and hence displacements of CM are identical on both sides. Now let left ladder go down by a distance x, then right ladder goes up by same distance. Now equating displacements of CM on both sides:

$$\frac{M(\ell - x) + 3M(-x)}{4M} = \frac{2M(x) + 2M(x-\ell)}{4M}$$

Solving we get $x = 3\ell/8$. Thus left ladder goes down by $3\ell/8$ dist.

EXAMPLE: PATHFINDER: BUILD 32:

A shell projected horizontally with a velocity of 165 m/s from the top of a 15 m high wall explodes into two identical fragments in the air. One of the fragments strikes the ground 1.5 s after the shell was projected at a distance of 240 m from the foot of the wall and 7 m to the right of the x-axis. Determine position of the other fragment at this instant. Neglect air resistance and assume acceleration of free fall to be 10 m/s^2.



Solution: Since explosion happens under action of internal forces, the CM continues to follow the same trajectory until one of the pieces falls to the ground. Assuming the pieces to be still in air, CM will fall at a time $T = \sqrt{2h/g} = \sqrt{3} = 1.732$ sec. Now since first piece lands at a time 1.5 sec, the other piece must still be in air.

Now $\vec{r}_{cm} = \frac{1}{2} g t^2 (-\hat{j}) + u t \hat{i} = -5 \times 1.5^2 \hat{j} + 165 \times 1.5 \hat{i}$

$+ 15\hat{j} \quad = 247.5\hat{i} - 11.25\hat{j} + 15\hat{j}$

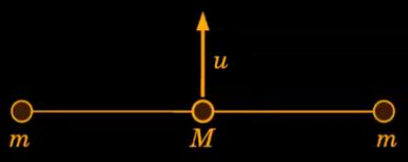
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Now landing point of first piece, $\vec{r}_1 = 240\hat{i} + 0\hat{j} + 7\hat{k}$

Now $\vec{r}_{cm} = \dfrac{m\vec{r}_1 + m\vec{r}_2}{2m} \Rightarrow \vec{r}_2 = 2(\vec{r}_{cm}) - \vec{r}_1$

$\Rightarrow \vec{r}_2 = 255\hat{i} + 7.5\hat{j} - 7\hat{k}$ Ans.

EXAMOCE: PATHFINDER: CHECK-3:

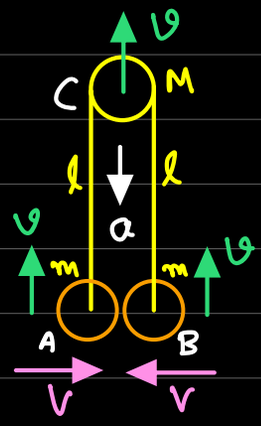


A disc of mass $M = 2.0$ kg is connected with two identical discs each of mass $m = 1.0$ kg with the help of two light inextensible threads each of lengths $l = 50$ cm. Initially the system rests on a frictionless horizontal floor with centres of the discs in a straight line and the threads straight. Now the middle disc is projected along the floor with a velocity $u = 2$ m/s perpendicular to the threads. Find tensile forces in the threads when the end discs are about to collide.

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Solution: Let the velocities of the discs along the thread just before collision be v and that of disc A and B perpendicular to strings be V. Then, using COM:

$$Mu = (M+2m)v \quad (1) \Rightarrow v = \left(\frac{M}{M+2m}\right)u$$



By COME: $\frac{1}{2}Mu^2 = \frac{1}{2}\left[(M+2m)v^2 + 2mV^2\right]$ (2)

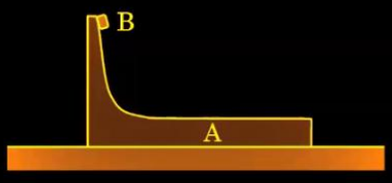
From (1) & (2), $V = \sqrt{\dfrac{M}{M+2m}}\,u$.

Now let acceleration of disc C be a as shown. Since there is no external force on the system, $\Sigma m_i a_i = 0$, thus: $-Ma + m\left(\frac{V^2}{l} - a\right)\times 2 = 0$

$\Rightarrow a = \dfrac{2mV^2}{l(M+2m)}$ Now $2T = Ma \Rightarrow T = \dfrac{Ma}{2}$

$\Rightarrow T = \dfrac{MmV^2}{l(M+2m)} = \dfrac{Mm}{l(M+2m)} \times \dfrac{M}{(M+2m)}u^2 = \dfrac{m}{l}\left[\dfrac{Mu}{M+2m}\right]^2$ Ans.

Example: Pathfinder MCQ 6:



Top of a large block A of mass 4 kg consists of a frictionless slope of height 40 cm which smoothly merges into a horizontal flat portion. The block rests on a frictionless horizontal floor. When a small block B of mass 1.0 kg is released on the top of the slope, it slides down the slope and then stops after sliding a distance 1.0 m on the horizontal portion of the top of the block A. Find coefficient of friction between the block B and horizontal portion of the top of the block A.

(a) 0.2 (b) 0.3
(c) 0.4 (d) 0.5

Solution: We know that work done by friction pair is independent of the frame, so we might as well find it in frame of A. Work done by friction = ΔME (CHIME). Thus:

$$\mu m_B g l = m_B g h \qquad (l: \text{sliding distance}, \mu: \text{friction coeff}).$$

Thus, $\mu = h/l = 40/100 = 0.4$ Ans (option C).

Example: Pathfinder: MCQ 8:



A small block rests on flat bottom of a deep bowl that is placed on a horizontal floor. There is no friction between the block and the bowl as well as between the bowl and the floor. Total mass of the bowl and the block is M. If the bowl is given a horizontal velocity by a sharp hit, the block rises to a maximum height h sliding on the walls of the bowl. Which of the following statements are correct?

(a) Maximum kinetic energy of the bowl is Mgh.
(b) If the bowl is lighter than the block, minimum kinetic energy of the bowl is zero.
(c) To predict minimum kinetic energy of the bowl, we must know ratio of the masses.
(d) To predict maximum as well as minimum kinetic energy of the bowl, we must know ratio of the masses.

Solution: Let us work from our Ninja Tehmik frame. When the block is at any of the extreme positions, $v_{cm} = 0$ from our frame $\therefore$ KE and momenta from CM frame are same as seen from our NTF. Now until the block reaches left extreme, Horizontal impulse of Normal reaction is μu where u is the relative velocity between block and the bowl. By CHIME:

$$\frac{1}{2}\mu u^2 = mgh \Rightarrow u^2 = \frac{2mgh}{\mu} \quad (1) \qquad (m: \text{mass of the block})$$

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Now during each rise or fall of the block, the magnitude of horizontal impulse on block is μu. Thus during the first half cycle of the motion, the bowl experiences a backward impulse of $2\mu u$ (μu during rise and μu during fall) and in second half cycle, it experiences a forward impulse of $2\mu u$ $\therefore$ the speed at the initial moment itself must be maximum (because thereafter it is being slowed down for the first half cycle).

Thus if m_0 is the mass of the bowl, its max KE,

$$KE_{max} = \frac{1}{2} m_0 u^2 = \frac{1}{2} m_0 \cdot \frac{2mgh}{\mu} = \frac{m_0 m g h}{m_0 m} (m_0 + m)$$

$\Rightarrow$ $KE_{max} = Mgh \Rightarrow$ (a) is correct.

(b) Now we need to work out whether or not Bowl reverses its direction. Max reverse impulse: $J_R = 2\mu u = \frac{2 m_0 m}{m_0 + m} u$,

while max forward momentum of Bowl is $m_0 u$, now if $m_0 < m$, $J_R / m_0 u = 2m/(m_0 + m) > 1$ i.e reverse impulse is more than initial forward momentum, $\therefore$ bowl must reverse its direction and $\therefore$ must pass through zero velocity.
$\therefore$ (b) is correct.

(c) To know minimum KE we need to know whether or not Bowl reverses its direction, which depends on ratio of masses
$\therefore$ to know min KE of the bowl, mass ratio will be required. (c) is correct.

(d) We already found max KE in part A, without knowing the ratio of masses. $\therefore$ d is incorrect. (Infact we could directly say that initial KE of the bowl must be maximum because Entire ME of the system at this instant is KE of the Bowl).
$\therefore$ d is not correct.

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Example: Pathfinder: MCQ 21-22:

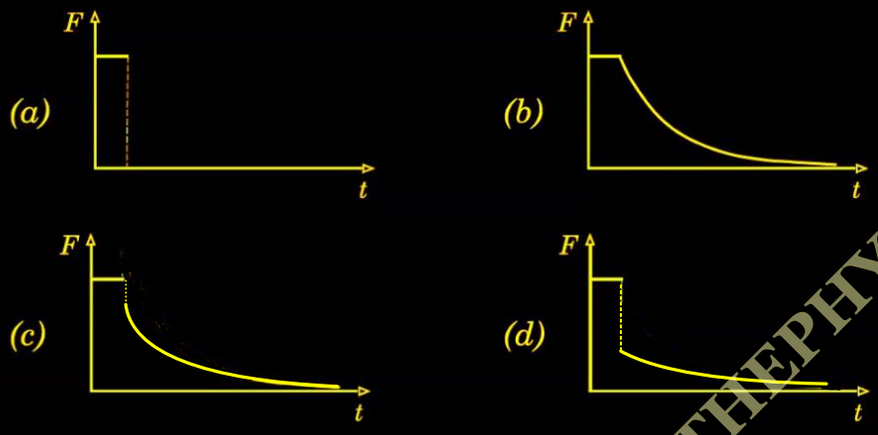


A plank of mass 1.5 kg is placed on a horizontal floor that is lubricated with oil. Top of the plank is in level with a platform as shown in the figure. Force of viscous drag on the plank due to layer of lubricating oil, when the plank slides is given by equation $\vec{f} = -k\vec{v}$, where $\vec{f}$ is in N, $\vec{v}$ is velocity of the plank in m/s and $k = 2.0$ kg/s. A small block of mass 0.5 kg lands from the platform on the plank with a velocity 10 m/s and after sliding some distance on the plank, the block stops on the plank.

21. How far does the plank slide on the floor?

(a) 1.0 m (b) 1.5 m
(c) 2.5 m (d) 3.0 m

22. Which of the following graph best describes relation between magnitude of frictional force F between the block and the plank with time t?



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Solution: (21): Using $\vec{F}_{ext} = \Sigma m_i \vec{a}_i$ we can write,

$\int \vec{F}_{ext}\, dt = \Sigma m_i \Delta\vec{v}_i$, thus using this equation for plank + block system, $\int -kv\, dt = m(0-u) + M(0-0)$

(∵ common velocity of block + plank = 0 finally), Here → m: mass of block, M: mass of the plank. Thus:

$-kS = -mu \Rightarrow S = mu/k = 0.5 \times 10/2 = 2.5$ m Ans. (option c).

(22) Before block comes to rest relative to the plank, friction is more than viscous drag because plank was accelerating forward (∴ $\mu mg > kv$). After block stops

relative to the plank, the plank starts decelerating ∴ friction must become smaller than kv, but not zero (∵ block is also decelerating with plank). ∴ (a) and (b) are ruled out.

Now let us see how to distinguish between (c) and (d):

Let f be the friction just after slipping stops, and let a be combined deceleration. Then:

$$kv = (M+m)a \quad ①$$

Now for block alone:

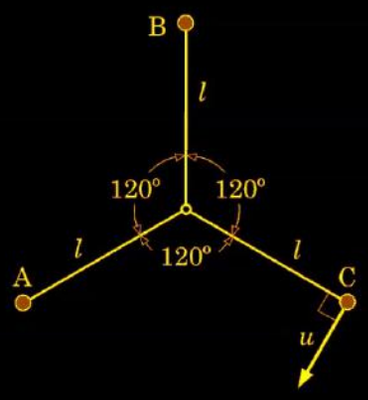
$$f = ma \quad ②$$

$$\Rightarrow \frac{f}{kv} = \left(\frac{m}{M+m}\right) = \frac{0.5}{1.5+0.5} = \frac{1}{4} \Rightarrow f = \frac{1}{4}kv.$$

Thus friction should drop to 1/4 times the value of viscous drag. This rules out option (c). The only correct option is (d).

EXAMPLE: PATHFINDER: MCQ 23-24:

Ends of three light identical rods each of length l are connected to a light pivot that enables them to rotate in any direction. At the other ends of the rods, three particles A, B and C of masses m, $2m$ and $3m$ respectively are affixed. Initially, the rods are coplanar, angle between any two adjacent rod is 120° and the particles are at rest. Now the particle C is given a velocity u perpendicular to the rod connected to it and in the plane of the rods as shown in the figure. Ignore gravitational interaction between the particles.



23:

Denoting acceleration vectors of the particles A, B and C by $\vec{a}_A$, $\vec{a}_B$ and $\vec{a}_C$ respectively, which of the following conclusion can you make?

(a) $\vec{a}_A + \vec{a}_B + \vec{a}_C = \vec{0}$ (b) $3\vec{a}_A + 2\vec{a}_B + \vec{a}_C = \vec{0}$

(c) $\vec{a}_A + 2\vec{a}_B + 3\vec{a}_C = \vec{0}$ (d) None of these

24:

What are moduli of acceleration vectors of the particles A, B and C immediately after the particle C is given velocity u?

(a) All equal to $\frac{u^2}{l}$. (b) $\frac{3u^2}{l}$, $\frac{3u^2}{2l}$ and $\frac{u^2}{l}$

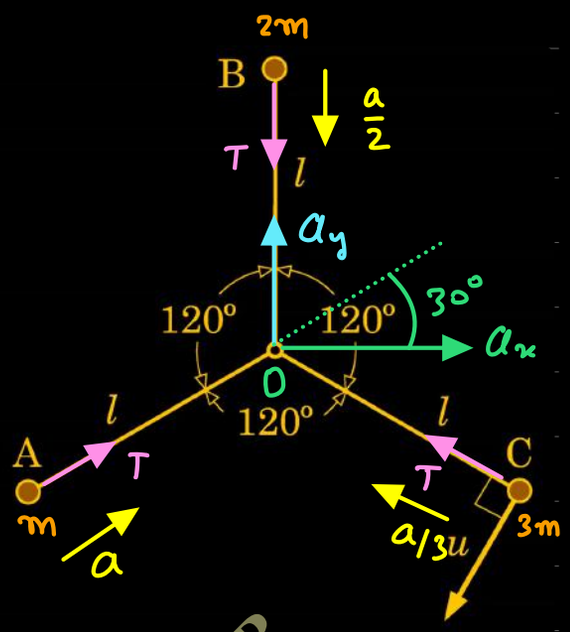
(c) $\frac{6u^2}{11l}$, $\frac{3u^2}{11l}$ and $\frac{2u^2}{11l}$ (d) $\frac{12u^2}{13l}$, $\frac{6u^2}{13l}$ and $\frac{4u^2}{13l}$

Solution: 23: Since there is no external force on the system ∴ acceleration of CM of the system must be zero. Thus:

$$m_A\vec{a}_A + m_B\vec{a}_B + m_C\vec{a}_C = 0 \Rightarrow m\vec{a}_A + 2m\vec{a}_B + 3m\vec{a}_C = 0 \Rightarrow$$ (C) is correct.

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(24) Since rods are massless, the tensions must be along the rods. Also since joint is massless, net force on it must be zero. Therefore by Lami's theorem, All the three tensions must be equal and accelerations must then be in the inverse ratio of masses. Let the acceleration components of the joint be a_x and a_y as shown in the figure. Now from joint frame, centripetal accelerations of A and B are zero and that of C is u^2/l. Writing the acceleration info in equation form:



$$a - a_x \cos 30° - a_y \sin 30° = 0 \quad (1) \quad \text{for A}$$

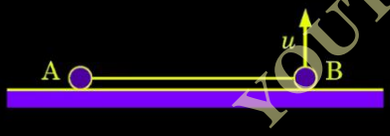
$$\frac{a}{2} + a_y = 0 \quad (2) \quad \text{for B and}$$

$$\frac{a}{3} + a_x \cos 30° - a_y \sin 30° = \frac{u^2}{l} \quad (3)$$

Solving (1), (2) & (3), we get: $a_x = 5\sqrt{3}u^2/l$, $a_y = -3u^2/11l$,

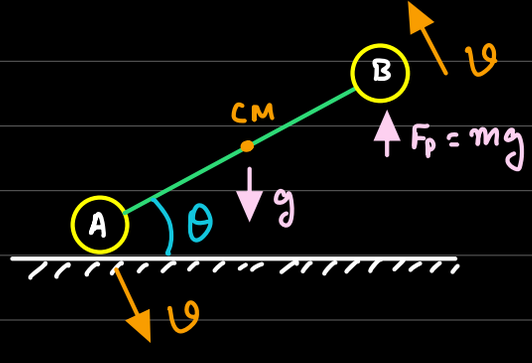
$a = 6u^2/11l$ Ans. Thus correct option is (C).

EXAMPLE: PATHFINDER: BUILD 13:



Two identical small balls A and B each of mass m connected by a light inextensible cord of length l are placed on a frictionless horizontal floor. With what velocity u must the ball B be projected vertically upwards so that the ball A leaves the floor? Acceleration of free fall is g.

<u>Solution:</u> Let contact be lost at an angle θ when speeds of the blocks in CM frame are as shown in the figure. At this instant acceleration of CM must be g downwards (because N is just zero). Now from FBD of the lower ball, from ground frame.



$T \sin\theta = mg$ (1) Note that from CM frame, gravity and pseudo force cancel each other. Also horizontal acceleration from CM frame must also be horizontal

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acceleration from ground frame. Thus, for lower ball:

$$\frac{m v^2}{(l/2)} \times \cos\theta = T\cos\theta \quad (2) \qquad \text{from } (1) \text{ \& } (2)$$

$$v^2 = gl/2\sin\theta, \quad (3)$$

Now Ball A has no vertical velocity relative to the ground (just lifting) If now V_{cm} is the vertical velocity of CM, then by constraint on A,

$$V_{cm} - v\cos\theta = 0 \Rightarrow V_{cm} = v\cos\theta.$$

Now total mechanical energy:

$$\frac{1}{2}(2m)V_{cm}^2 + \frac{1}{2}\mu(2v)^2 + mg\,l\sin\theta, \quad \text{substituting } v^2 \text{ from } (3) \text{ and putting } \mu = m/2,$$

we get:

$$E = mgl\left[\frac{\cos^2\theta}{2\sin\theta} + \frac{1}{2\sin\theta} + \sin\theta\right] \quad \text{and we need}$$

to minimize this wrt θ. Writing $\cos\theta = 1-\sin^2\theta$, we must minimize $mgl\left[\frac{1}{\sin\theta} + \frac{\sin\theta}{2}\right]$, thus, $-\text{cosec}\,\theta\cot\theta + \frac{\cos\theta}{2} = 0$

$$\Rightarrow \cos\theta\left[\frac{-1}{2} + \frac{1}{\sin^2\theta}\right] = 0 \Rightarrow \theta = \pi/2 \quad \text{accordingly, min Energy:}$$

$$E_{min} = \frac{3}{2}mgl = \frac{1}{2}mu^2 \quad (\text{where } u \text{ is the required velocity of the ball})$$

thus $u_{min} = \sqrt{3gl}$ Ans.

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